



Jeremías Likerman

ADJUNCT RESEARCHER - CONICET · ASSOCIATE PROFESSOR - UNIVERSITY OF BUENOS AIRES
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Profile

Research profile: Geologist specializing in structural geology, geomechanics, and multiscale numerical modelling, with more than 15 years of experience in research, university teaching, fieldwork, and student training. His research integrates two- and three-dimensional models to study geodynamic, geothermal, and geomechanical processes in the Andes and in energy reservoirs, including subduction zones, crustal deformation, volcanic systems with geothermal potential, and natural and induced fracturing, with emphasis on the Vaca Muerta Formation.

Teaching, knowledge transfer, and management: Associate Professor at the University of Buenos Aires and Adjunct Researcher at CONICET. His university teaching career began in 2013 as a First-Class Teaching Assistant, continued as Head Teaching Assistant from 2018 to 2026, and has progressed to Associate Professor since 2026. He has taught Structural Geology, Geotectonics, Field Tectonics, Introduction to Geology, Geological Surveying, Geostatistics, and Geoinformatics; supervises and co-supervises researchers and students; coordinates national and international projects; and participates in knowledge-transfer, public science communication, and institutional management activities at IDEAN.

Academic and Postdoctoral Training

Postdoctoral Fellowship

Argentina / Spain

CONICET AND UNIVERSITAT POLITÈCNICA DE CATALUNYA

2015–2017

- Topic: thermal evolution of oceanic lithosphere. Supervisors: Dr. Ernesto Cristallini (CONICET–UBA) and Dr. Sergio Zlotnik (UPC).

Visiting Researcher

United States

STANFORD UNIVERSITY

2013

- Courses in Structural Geology and MATLAB applied to Geosciences, as part of a Fulbright–Bunge and Born Foundation fellowship.

PhD in Geological Sciences

Argentina

UNIVERSITY OF BUENOS AIRES

2010–2015

- Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Dr. Ernesto Cristallini.

BSc in Geological Sciences

Argentina

UNIVERSITY OF BUENOS AIRES

2005–2010

- Thesis: *Geology and structure of the Tres Cruces compound anticline, Jujuy, Argentina*. Supervisor: Dr. Ernesto Cristallini.

Appointments and Professional Experience

CURRENT APPOINTMENTS

Adjunct Researcher

Argentina

CONICET–IDEAN

2018–present

- Research career since 2018; promoted to Adjunct Researcher in November 2023. Research lines: fracture prediction, geodynamic modelling, and numerical assessment of geothermal potential.

Associate Professor

Argentina

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

June 2026–present

- Courses: Geostatistics, Geoinformatics, and Geological Surveying. Full-time appointment.

Institutional Coordinator

Argentina

IDEAN, CONICET–UBA

September 2025–present

- Member of the institute council since April 2024.

PREVIOUS TEACHING APPOINTMENTS

Head Teaching Assistant (Jefe de Trabajos Prácticos)

Argentina

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

March 2018–May 2026

- Courses: Introduction to Geology, Geological Surveying, and Geostatistics.

First-Class Teaching Assistant (Ayudante de Primera)

Argentina

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

June 2013–September 2017

- Courses: Structural Geology, Geotectonics, and Field Tectonics.

COURSES AND FIELD ACTIVITIES

Geology of Unconventional Reservoirs

Argentina

INSTRUCTOR IN CHARGE

2025

- Professional training course focused on the geological characterization of shale reservoirs.

Advanced Research Field Trip on Vaca Muerta

Argentina

ORGANIZER AND INSTRUCTOR

2025

- *Perspectives on Vaca Muerta: subsurface attributes analysed from surface exposures.*

RESEARCH VISITS

Visiting Researcher

Spain

GEO3BCN-CSIC

May–September 2025

- Thermomechanical modelling of volcanic zones for geothermal-potential assessment. Host: Dr. Ivone Jiménez-Munt.

Visiting Researcher

Spain

GEO3BCN-CSIC

August–September 2024

- Development of coupled numerical models for volcanic and geothermal systems.

Visiting Researcher

Spain

BARCELONATECH (UPC)

2015, 2017, 2019, and 2022

- Research stays at the Numerical Computation Laboratory.

Research Projects

2026	Three-dimensional models of stress transfer between adjacent Andean flat-slab segments (Principal Investigator). AECT-2026-1-0080	Spain
2026–2029	Tectonic-structural analysis through analogue and numerical modelling: contributions to energy resources (Co-Principal Investigator). PIP 11220250100817CO	Argentina
2025	Geothermal potential of volcanic zones (Principal Investigator). AECT-2025-1-0009	Spain
2025–2026	Structural controls in volcanic systems: magmatic intrusions and damage-zone development associated with hydrocarbon generation (Co-Principal Investigator). Williams Foundation	Argentina
2024–2026	Geothermal-potential estimation in volcanic zones through thermomechanical numerical models (Principal Investigator). LINCGL2024	Spain
2024	Geodynamic modelling of subduction zones and mantle plumes (Principal Investigator). AECT-2024-2-0027	Spain
2023	Geodynamics of the Adria microplate, from mantle to surface (Co-Principal Investigator). GEO3BCN-CSIC	Spain
2022	Geodynamic modelling of subduction zones (Principal Investigator). AECT-2022-3-0023	Spain

2020	Landscape dynamics and geological hazards associated with dams in mountain regions (Co-Principal Investigator). PICT-2019-04011	Argentina
2020	Analogue and numerical modelling: contributions to tectonic and structural problems (Co-Principal Investigator). PICT-2019-00997	Argentina
2018	Natural fracturing and first-order structures: predictive models applied to Cerro Domuyo (Co-Principal Investigator). UBACyT	Argentina
2017	Geological evolution of the Andes and its economic and environmental impact (Co-Principal Investigator). PUE 22920160100051	Argentina
2016	Numerical modelling of the thermal evolution of oceanic lithosphere (Principal Investigator). PICT 2015-1226	Argentina
2016	Application of analogue and numerical models to tectonic-structural problems (Co-Principal Investigator). PICT-2016-1407	Argentina

Research Supervision

RESEARCHER SUPERVISION

Plotek, Berenice

ASSISTANT RESEARCHER

Argentina

Since August 2025

- **Supervisor:** Likerman, J. **Co-supervisor:** Cristallini, E.

POSTDOCTORAL SUPERVISION

Plotek, Berenice

Argentina

FAULT-PROPAGATION FOLDS: A MECHANICAL APPROACH USING FINITE-ELEMENT NUMERICAL MODELLING

2024

- **Supervision:** Cristallini, E. and Likerman, J.

DOCTORAL THESIS SUPERVISION

Correa Luna, Clara

Argentina

NATURAL FRACTURING IN THE VACA MUERTA FORMATION: CHARACTERIZATION AND MODELLING

2023

- **Supervision:** Yagupsky, D. and Likerman, J.

UNDERGRADUATE THESIS SUPERVISION

Chandía Sepúlveda, Alexander Ignacio

Argentina

TECTONIC INVERSION OF A METAMORPHIC COMPLEX IN THE DESEADO MASSIF: CONTRIBUTIONS FROM ANALOGUE MODELLING

In progress

- **Supervision:** Likerman, J. and Navarrete, C.

De Rosa, Tomás

Argentina

QUATERNARY DEFORMATION IN THE FIAMBALÁ VALLEY, CATAMARCA

In progress

- **Supervision:** Likerman, J. and Sagripanti, L.

Blanco, Lucía

Argentina

ASSESSMENT OF THE GEOTHERMAL-ELECTRIC POTENTIAL OF THE CERRO BLANCO VOLCANIC COMPLEX, CATAMARCA: INFLUENCE OF RHEOLOGY AND STRUCTURE THROUGH NUMERICAL MODELS

In progress

- **Supervision:** Likerman, J. and Lamberti, C.

Celis Santisteban, Celso Angel

Argentina

STRUCTURAL EVOLUTION OF AN INVERTED METAMORPHIC COMPLEX IN THE DESEADO MASSIF: CONTRIBUTIONS FROM NUMERICAL MODELLING

In progress

• Supervision: Likerman, J. and Navarrete, C.	
Ruiz, Luciana	Argentina
ASSESSMENT OF THE GEOTHERMAL-ELECTRIC POTENTIAL OF THE CERRO BLANCO VOLCANIC COMPLEX, CATAMARCA, THROUGH NUMERICAL MODELLING	2026
• Supervision: Lamberti, C. and Likerman, J.	
Gramajo, Pilar	Argentina
QUATERNARY DEFORMATION IN THE CHASCHUIL VALLEY, WESTERN SIERRA DE LAS PLANCHADAS, CATAMARCA	2025
• Supervision: Likerman, J. and Sagripanti, L.	
Acosta, Luana	Argentina
GEOMORPHOLOGY OF THE RÍO GRANDE VALLEY BETWEEN BARDAS BLANCAS AND PORTEZUELO DEL VIENTO, MENDOZA	2023
• Supervision: Winocur, D. and Likerman, J.	
Mazzini, Martín	Argentina
GEOLOGY OF THE EASTERN SLOPE OF CERRO PERITO MORENO, EL BOLSÓN, RÍO NEGRO	2022
• Supervision: Likerman, J. and Tobal, J.	
Relañez, Gastón	Argentina
STRUCTURE AND NEOTECTONICS OF THE TABLADITAS AREA, JUJUY, ARGENTINA	2019
• Supervision: Yagupsky, D. and Likerman, J.	
Ventisky, Esteban	Argentina
GEOLOGY, STRUCTURE, AND FRACTURING OF THE LAJAS FORMATION IN THE COVUNCO ANTICLINE, NEUQUÉN	2017
• Supervision: Likerman, J. and Tobal, J.	
Correa Luna, Clara	Argentina
STRUCTURAL GEOLOGY AND FRACTURING IN THE TRES CRUCES BASIN, JUJUY	2015
• Supervision: Yagupsky, D. and Likerman, J.	

Fellowships and Distinctions

2027	Research Fellowship in Japan Kobe University	JSPS
2025	SZNet 2025 Chile Field Trip National Science Foundation	SZ4D
2022	Academic Mobility Fellowship among Partner Institutions Ibero-American University Association for Postgraduate Studies	AUIP
2016	National Scientific and Technical Research Council International Research Stay	CONICET
2015	National Scientific and Technical Research Council Postdoctoral Fellowship	CONICET
2013	Stanford University Fulbright – Bunge & Born Foundation	Doctoral fellowship
2013	National Scientific and Technical Research Council Doctoral Fellowship Type II	CONICET
2010	National Scientific and Technical Research Council Doctoral Fellowship Type I	CONICET

Skills

Programming and reproducibility: $\text{\LaTeX}$, Python, R, MATLAB, HTML5, Git, and reproducible scientific documentation.

Numerical modelling and HPC: Underworld2, ASPECT, GOLEM, and LaMEM; MPI, post-processing, and visualization of 2D/3D models.

Analogue modelling: experimental design, material characteriza-

tion, PIV, and kinematic and mechanical analysis.

Spatial and geological analysis: QGIS, GMT, and PostGIS; geological mapping, seismic interpretation, structural analysis, and UAV-assisted field methods.

Fieldwork: more than 15 years of experience in the Andes; campaign

planning, logistics, and team leadership.

Languages: Spanish native, English fluent (C2), Italian (A2), and Catalan (A2).

Peer-Reviewed Publications

1. Analogue modelling to enhance spatial reasoning and understanding of tectonic processes in geoscience education. Jara, P. P., Pinto, L., **Likerman, J.** *Andean Geology* 53.3 (2026)
2. The role of compressional structures in the development of collapse calderas: Insights from analogue models. Villarroel, M. A., Browning, J., Jara, P. P., Harnett, C. E., Holohan, E. P., Marquardt, C. J., Guzman, M., **Likerman, J.** *Journal of Volcanology and Geothermal Research* (2026) p. 108582
3. Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén basin, Argentina. Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H. *Tectonophysics* (2025) p. 230640
4. Slab underthrusting is the primary control on flat slab size. Gianni, G. M., Gallo, L. C., **Likerman, J.**, Echaurren, A., Gianni, C. R., Faccena, C. *Science Advances* 11.28 (2025)
5. Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., Frassetto, M. P. *Earth-Science Reviews* 249 (2024) p. 104651
6. Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. Plotek, B., **Likerman, J.**, Cristallini, E. *Journal of Structural Geology* (2024) p. 105064
7. Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction. Gianni, G. M., **Likerman, J.**, Navarrete, C. R., Gianni, C. R., Zlotnik, S. *Nature communications* 14.1 (2023) p. 2083
8. Natural fracture characterization of the Los Catutos Member (Vaca Muerta Formation) in the southern sector of the Sierra de la Vaca Muerta, Neuquén Basin, Argentina. Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H. *Journal of South American Earth Sciences* 120 (2022) p. 104081
9. Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations. Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., **Likerman, J.** *Journal of Structural Geology* 162 (2022) p. 104703
10. The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes. García Morabito, E., Beltrán-Triviño, A., Terrizzano, C. M., Bechis, F., **Likerman, J.**, Von Quadt, A., Ramos, V. A. *Tectonics* 40.2 (2021) e2020TC006374
11. The effects of small-scale convection in the shallow lithosphere of the North Atlantic. **Likerman, J.**, Zlotnik, S., Li, C.-F. *Geophysical Journal International* 227.3 (2021) pp. 1512–1522
12. Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina. Luna, C. C., Yagupsky, D. L., **Likerman, J.** *Journal of South American Earth Sciences* (2019) p. 102446
13. Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate. Gianni, G. M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H. P. A., Dal Molin, C., Valencia, V. A. *Tectonophysics* 721 (2017) pp. 151–166
14. Closure type effects on the structural pattern of an inverted extensional basin of variable width: Results from analogue models. Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., Winocur, D. *Journal of South American Earth Sciences* (2017)
15. Climatic and Tectonic forcing on alluvial fans in the Southern Central Andes. Terrizzano, C. M., Morabito, E. G., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., Zech, R. *Quaternary science reviews* 172 (2017) pp. 131–141
16. Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory: Implications for their effects on groundwater flow and contaminant transport. Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., Cherry, J. *Journal of Structural Geology* 85 (2016) pp. 95–114
17. Middle to late Miocene extensional collapse of the North Patagonian Andes (41° 30'–42° S). Tobal, J. E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F. L., Ramos, V. A., Gimenez, M. *Tectonophysics* 657 (2015) pp. 155–171
18. Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes. Ghiglione, M. C., **Likerman, J.**, Barberón, V., Giambiagi, L. B., Aguirre-Urreta, B., Suarez, F. *Basin Research* 26.6 (2014) pp. 726–745
19. Role of basin width variation in tectonic inversion: insight from analogue modelling and implications for the tectonic inversion of the Abanico Basin, 32°–34° S, Central Andes. Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M. C., Cristallini, E. O., Pinto, L., Charrier, R. *Geological Society, London, Special Publications* 399 (2014) SP399–7
20. Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling. **Likerman, J.**, Burlando, J. F., Cristallini, E. O., Ghiglione, M. C. *Tectonophysics* 590 (2013) pp. 106–120